

PROJECT BUTTON HR–V0 ELECTRICAL V3 CONNECTED CANDIDATE

Separate RESET and ARM, two PNOZ stages, dual watchdog contacts, external adapters, redundant actuator interruption.

01 External listed sources and DC boundaries

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File: 08_compute_and_control_terminals.kicad_sch

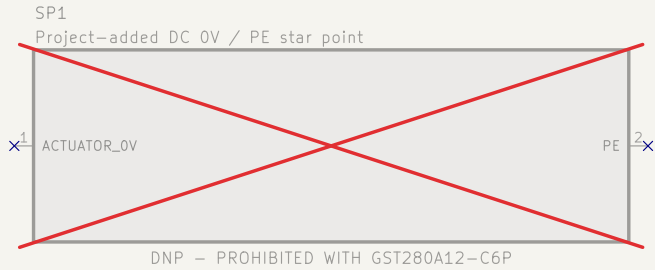
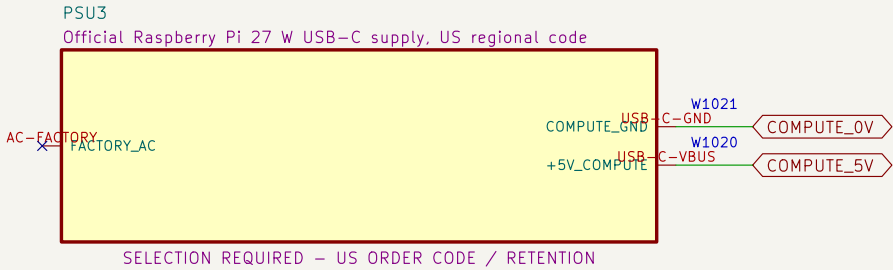
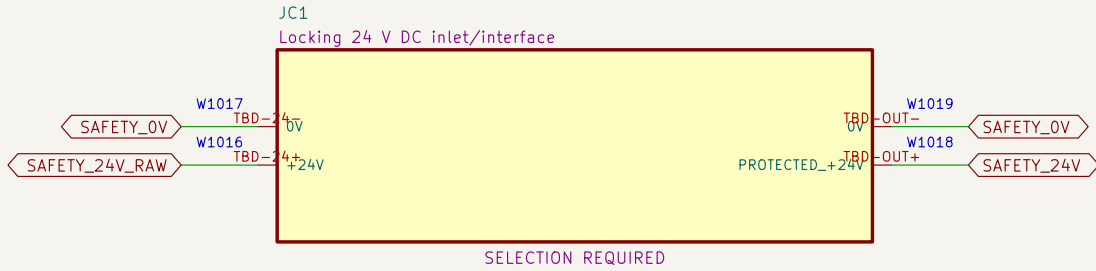
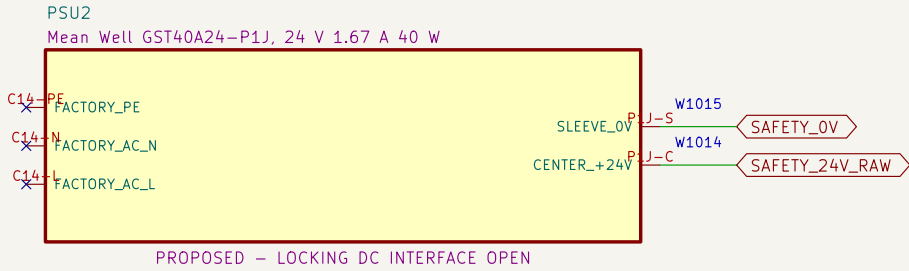
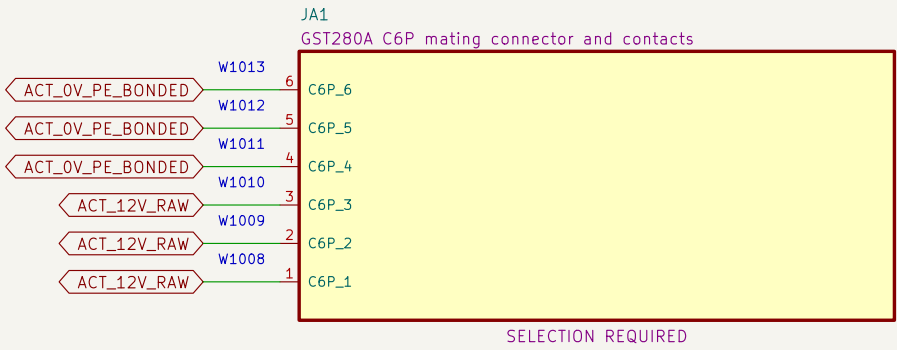
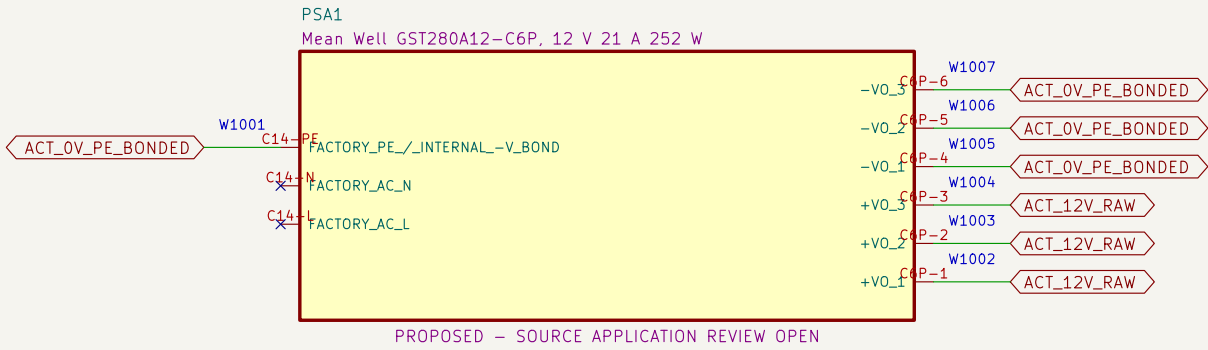
09 U2D2, actuator ports and bonding boundary

File: 09_actuator_interfaces.kicad_sch

V2.1 PRESERVED; V3 IS A CONNECTED CANDIDATE PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION Project Button			
Sheet: / File: project–button–v3.kicad_sch			
Title: Project Button HR–V0 Electrical V3 index			
Size: A3		Date: 2026–08–06	Rev: V3–P0.1
KiCad E.D.A. 10.0.5			Id: 1/10

01 External listed sources and DC boundaries

Factory-sealed adapters eliminate project-built mains wiring.



NOTE 1: All AC conductors shown are factory boundaries, not project-built wiring.

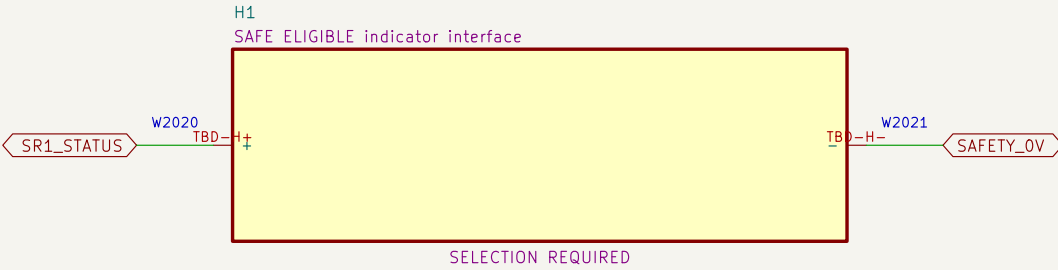
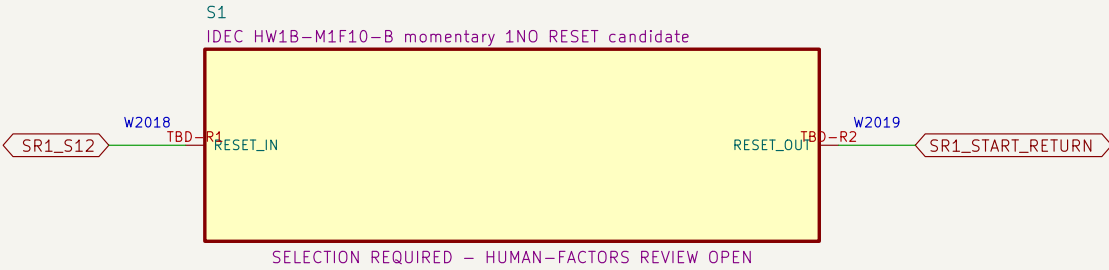
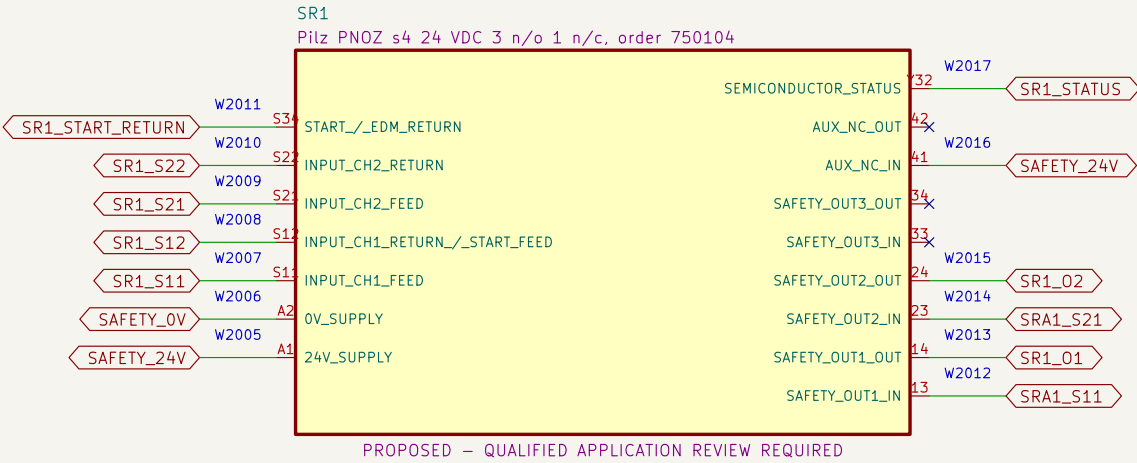
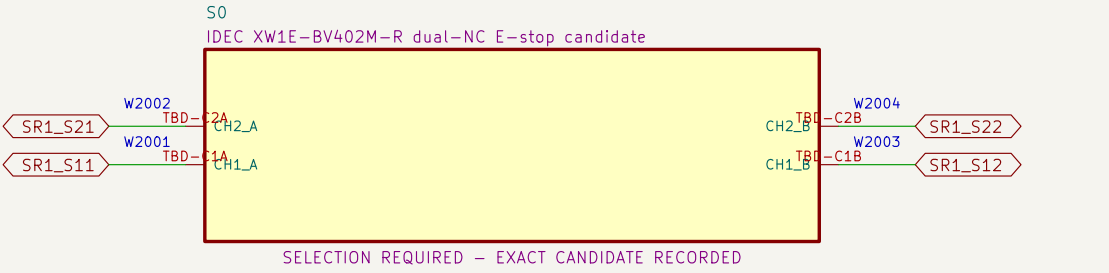
NOTE 2: Site cords, receptacles, GFCI/code basis and source application review remain open.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /01 External listed sources and DC boundaries/
File: 01_external_sources.kicad_sch

Title: PB HR–V0 Electrical V3 – 01
Size: A3 Date: 2026–08–06 Rev: V3–P0.1
KiCad E.D.A. 10.0.5 Id: 2/10

02 Dual-channel E-stop and RESET eligibility

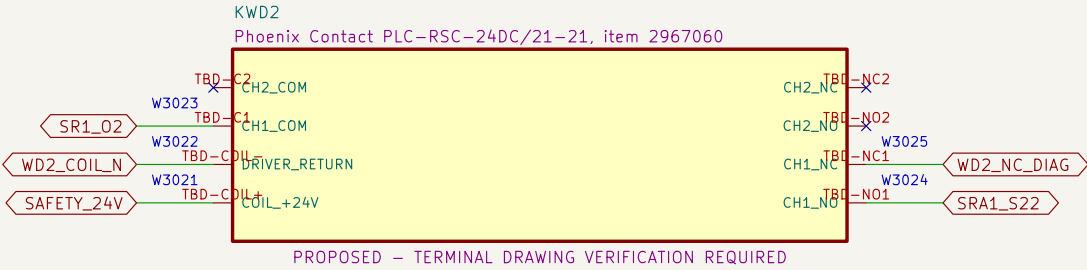
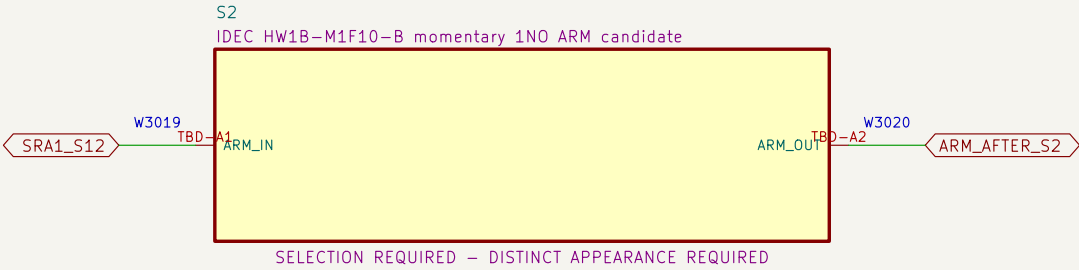
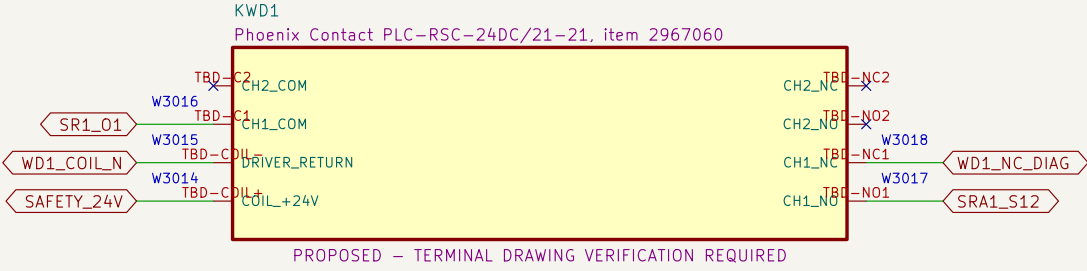
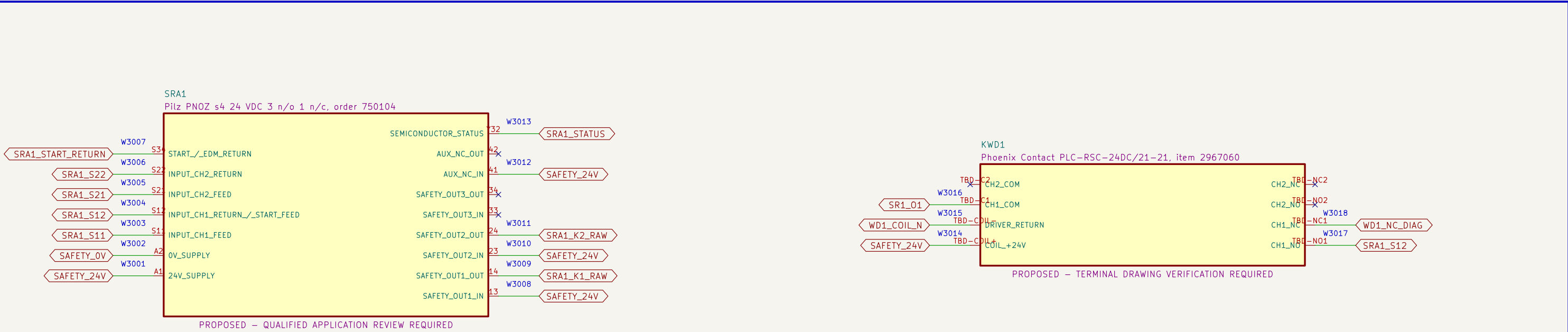
SR1 monitors E-stop channels. RESET cannot energize the actuator contactors.



NOTE 1: RESET release may make SR1 eligible, but SRA1 and K1/K2 remain de-energized.
NOTE 2: Unused SR1 outputs are explicitly named; do not bridge them during wiring.

03 Distinct ARM and watchdog eligibility

SRA1 requires SR1 eligibility, two watchdog channels, EDM proof and a new ARM action.

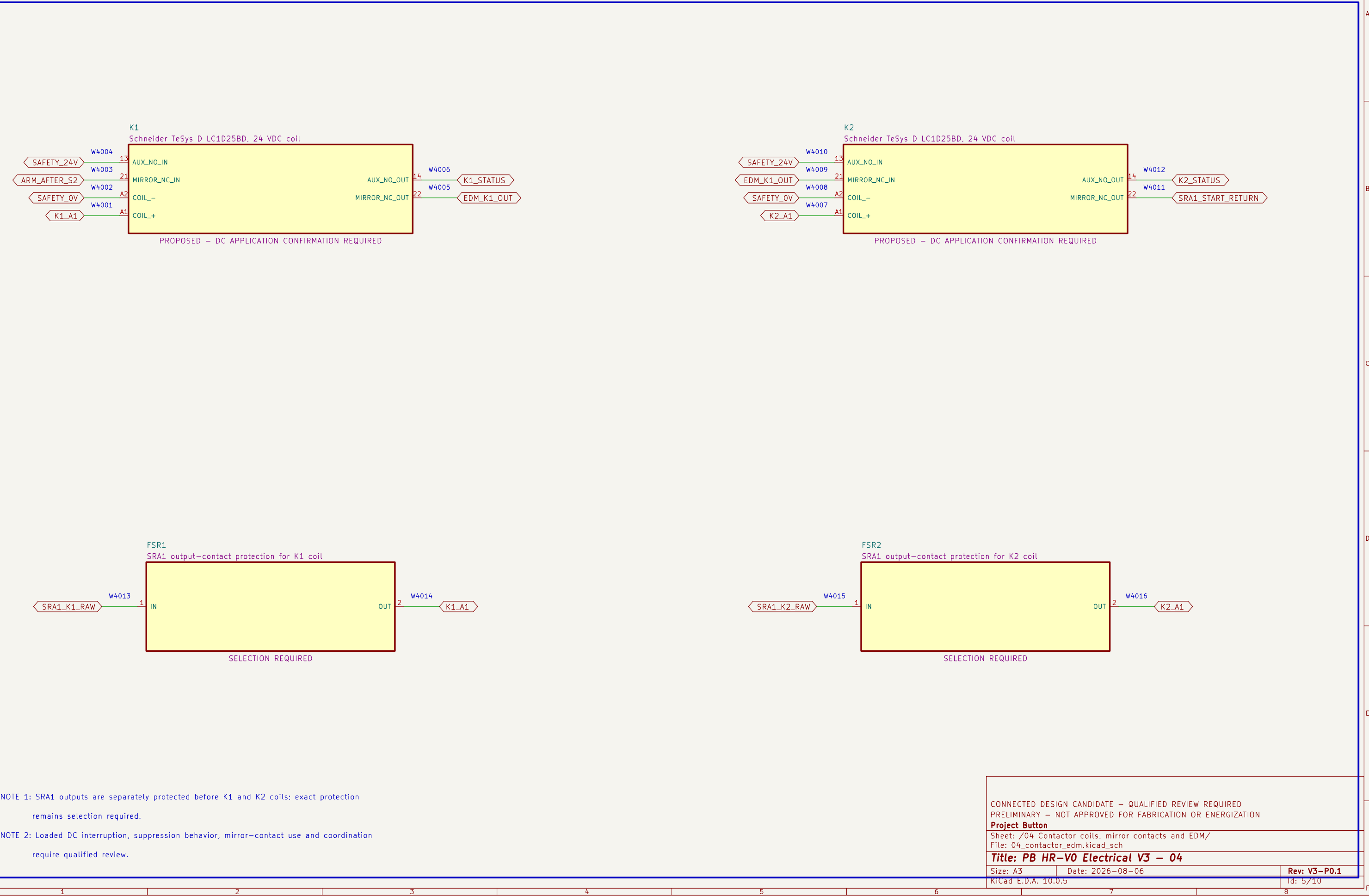


NOTE 1: Required sequence: E-stop healthy -> RESET press/release -> SAFE_READY -> distinct
ARM press/release.

NOTE 2: Heartbeat restoration alone cannot reclose SRA1; a monitored ARM action is required
after dropout.

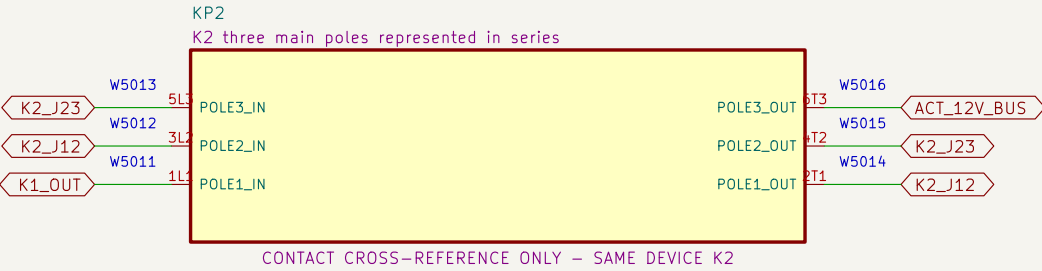
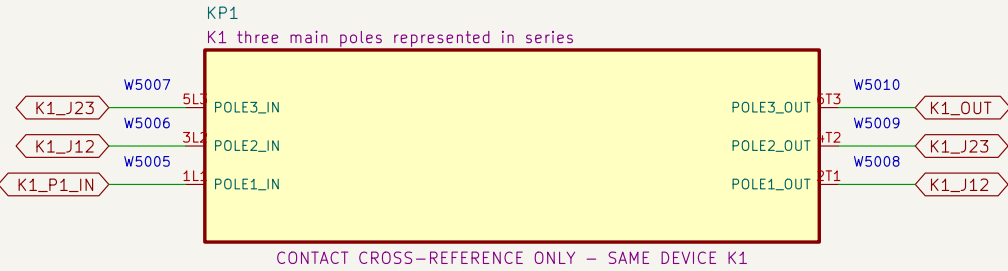
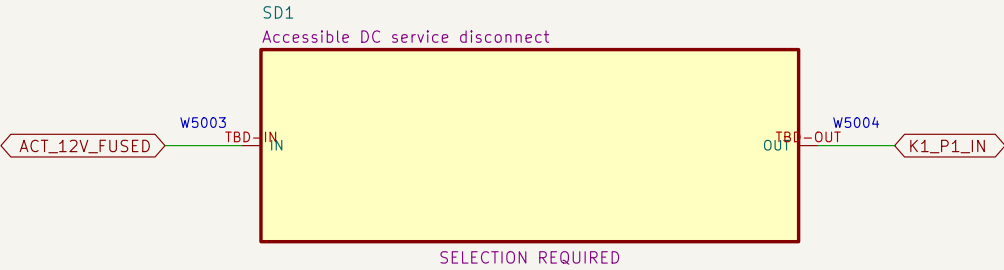
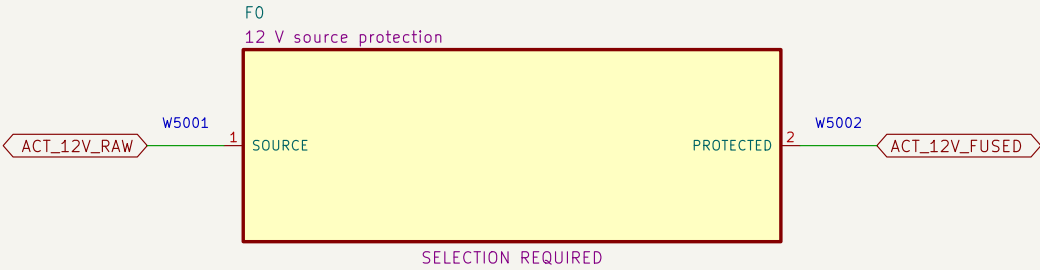
04 Contactor coils, mirror contacts and EDM

K1 and K2 are distinct final elements; their mirror contacts form the monitored restart return.



05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



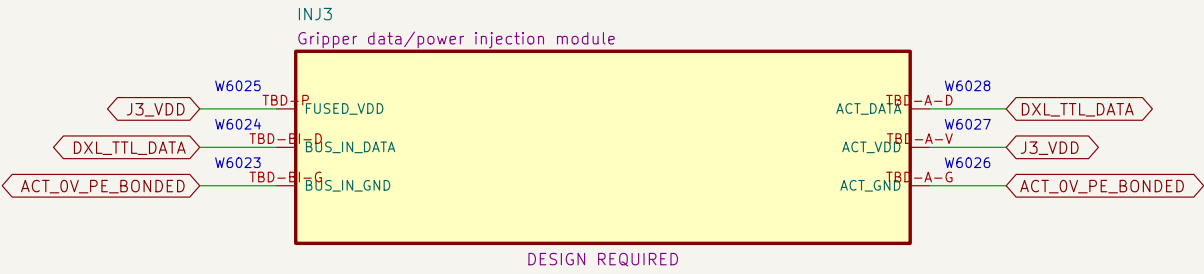
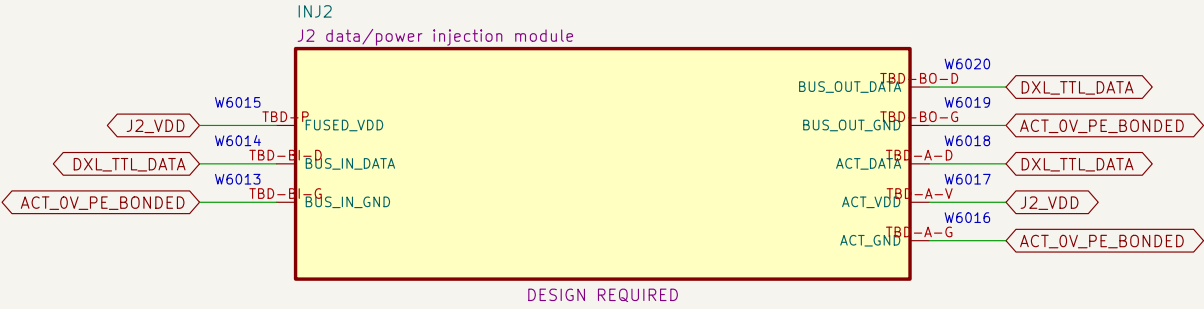
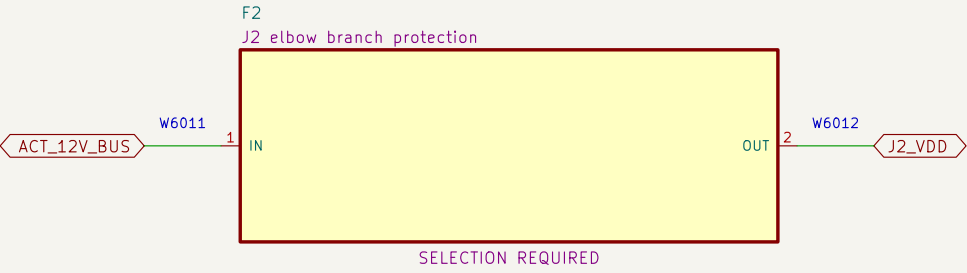
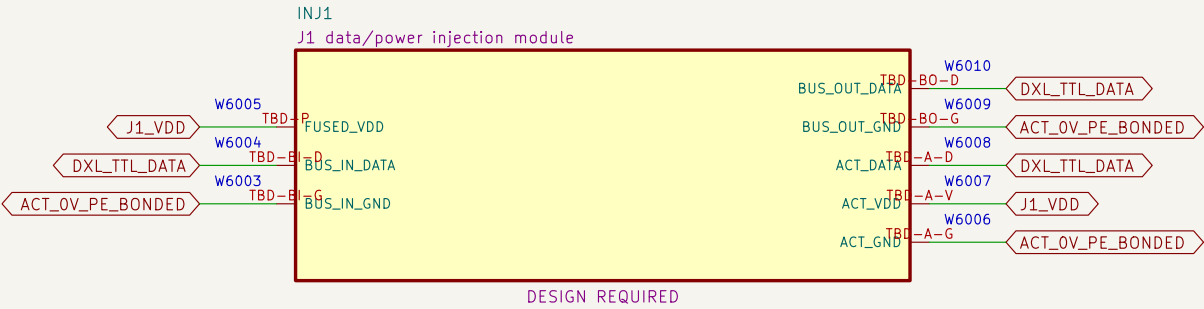
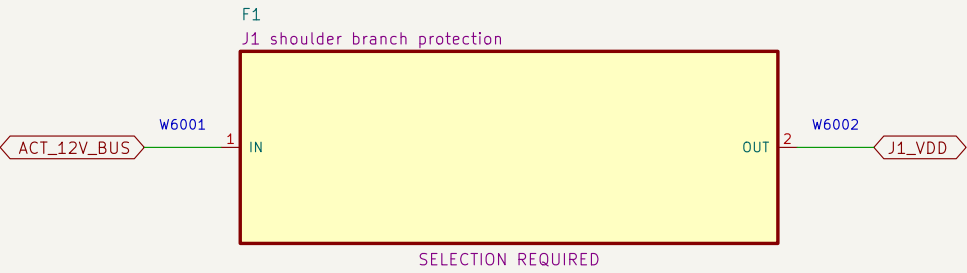
NOTE 1: Series jumpers: K1 2T1→3L2, 4T2→5L3, 6T3→K2 1L1; repeat through K2 to ACT_12V_BUS.

NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault–current evidence.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED		
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION		
Project Button		
Sheet: /05 Redundant actuator–power interruption/ File: 05_actuator_interruption.kicad_sch		
Title: PB HR–V0 Electrical V3 – 05		
Size: A3	Date: 2026–08–06	Rev: V3–P0.1
KiCad E.D.A. 10.0.5		Id: 6/10

06 Protected actuator branches and VDD-isolating injection

Each actuator has a separate protected VDD branch; inter-actuator data cables must omit VDD.



NOTE 1: Standard fully populated ROBOTIS TTL daisy-chain cables are prohibited because they would parallel branch VDD.

NOTE 2: Every injection module requires released source, continuity, isolation, pull and no-backfeed tests.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

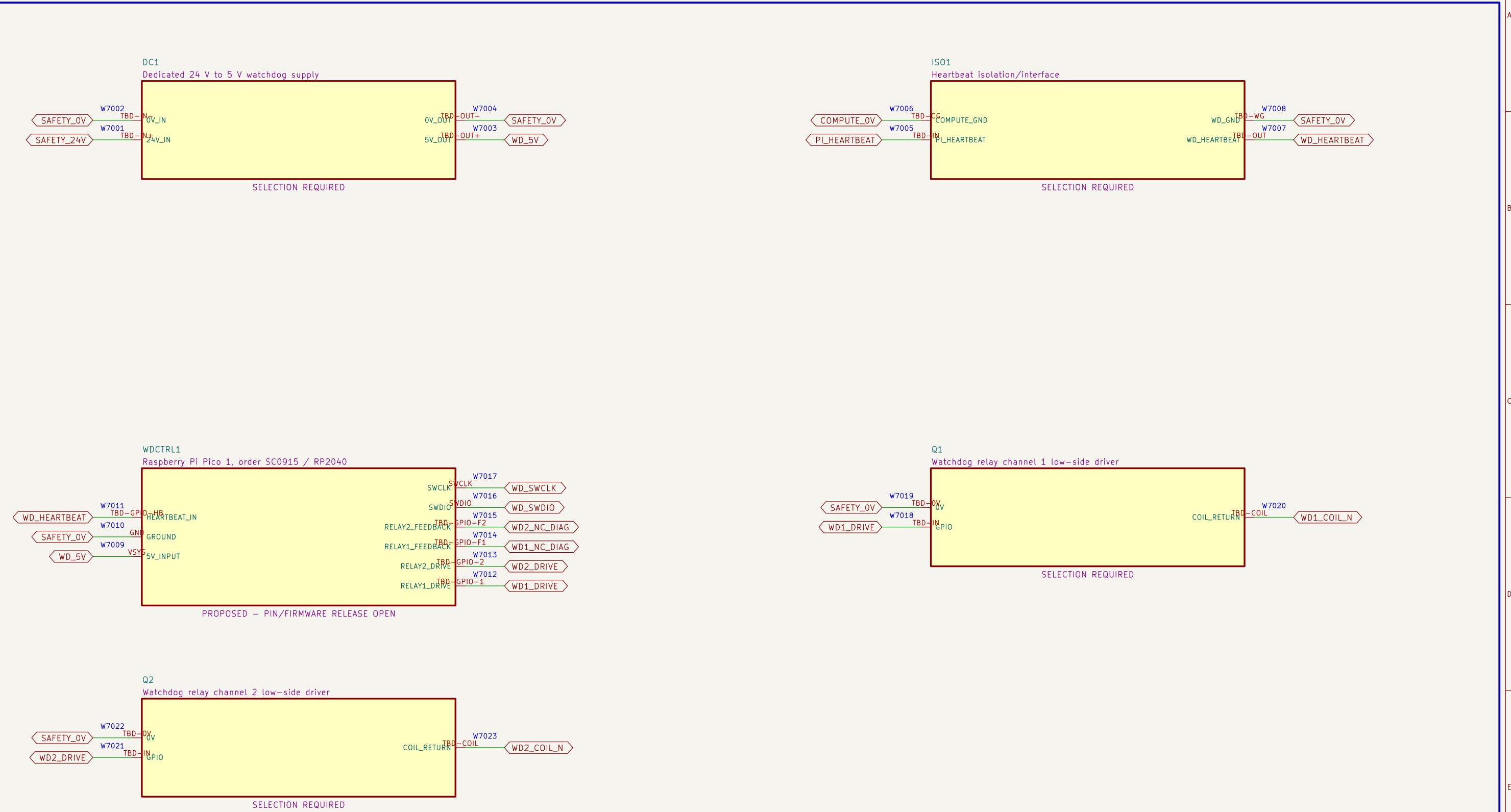
Project Button
Sheet: /06 Protected actuator branches and VDD-isolating injection/
File: 06_branches_and_injection.kicad_sch

Title: PB HR-V0 Electrical V3 – 06

Size: A3	Date: 2026-08-06	Rev: V3-P0.1
KiCad E.D.A. 10.0.5		Id: 7/10

07 Independent watchdog power, controller and drivers

The watchdog forces restart but receives no safety–integrity credit by assertion.

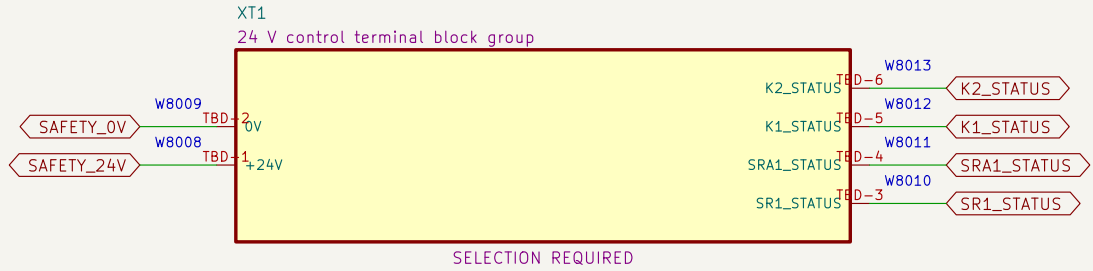
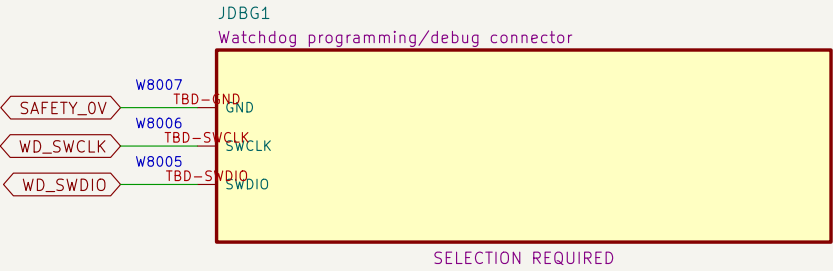
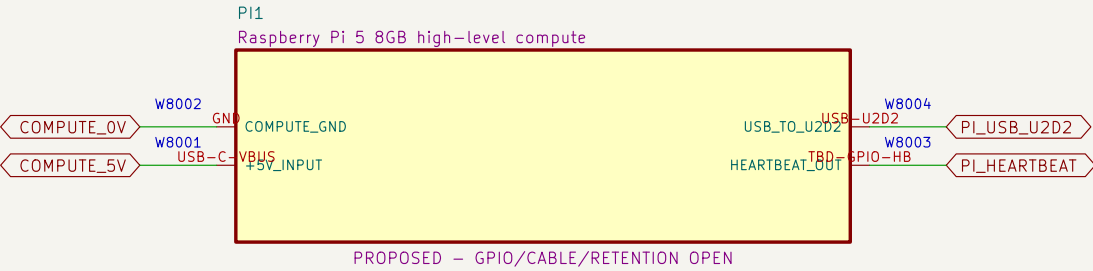


NOTE 1: Power–up, brownout, clock failure, stuck GPIO, held heartbeat and firmware–corruption tests are mandatory.

NOTE 2: A qualified review must decide whether watchdog loss is credited or diagnostic only.

08 Compute, debug and control terminals

High-level compute and diagnostic wiring have no authority to bypass or restore the safety chain.



NOTE 1: Programming/debug connections must not enable outputs during operation.

NOTE 2: Terminal family, markers, conductor range, torque and enclosure layout remain selection required.

09 U2D2, actuator ports and bonding boundary

The U2D2 cable carries DATA and GND only; protected VDD is injected at each actuator.

